Important Topics from Your SQL Notes

Prioritised for interviews, with the exact points to say

Priority key: Tier 1 = asked in almost every interview. Tier 2 = very likely. Tier 3 = in your notes, often asked by experienced interviewers (normalization, dependencies, attribute types). Based on a reading of your handwritten notes; if a point in your own page differs, follow your teacher's version.

TIER 1: Must know

1. DBMS vs RDBMS vs NDBMS (comparison table in your notes)

Point	RDBMS	NDBMS (No-SQL)
Storage format	Table format (structured)	File format (unstructured)
Language	SQL	No SQL
Handling	Very easy	Difficult
Data size	Small to medium datasets	Large scale datasets
CRUD speed	Takes less time	Takes more time
Best for	Text-related information	Multimedia information
Schema validation	Mandatory (fixed schema)	Optional (no fixed structure)
Properties	ACID (Atomicity, Consistency, Isolation, Durability)	ACID and CAP (Consistency, Availability, Partition tolerance)
Scaling	Vertical only	Vertical and horizontal
Relations	Supports joins and sub queries	No joins or sub queries
Examples	Oracle, MySQL, PostgreSQL, MS SQL, T-SQL	MongoDB, HBase, Amazon DynamoDB, Firebase

Say it simply: DBMS stores data as tables or files. If tables, it is an RDBMS (structured, SQL). If files, it is a NDBMS (non-structured, No-SQL). NDBMS types: network, object-oriented, hierarchy based.

2. Datatypes

Datatype	Points to say
CHAR(size)	Fixed length, max 2000. Unused memory is wasted. Good for fixed data (PAN, pincode).
VARCHAR / VARCHAR2(size)	Variable length, max 4000 for VARCHAR2. Unused memory is returned, so it saves space. VARCHAR2 is the updated version.
DATE	Stores dates in Oracle format.
NUMBER(p,s)	Numbers with precision and scale.
CLOB / BLOB	Large objects up to 4 GB. Character large object (text) and Binary large object (images, files).

3. Constraints and schema validation

Schema validation = checking that the proper data enters the column, done with datatypes and constraints (rules). A table is also called a schema.

Constraint	Rule	Syntax
NOT NULL	Value is mandatory	col NUMBER NOT NULL
UNIQUE	No duplicates, NULL allowed	col VARCHAR2(30) UNIQUE
PRIMARY KEY	Unique + not null, one per table, parent table	col NUMBER PRIMARY KEY
FOREIGN KEY	Refers to PK of another table, child table	col NUMBER REFERENCES parent(col)
CHECK	Condition must be true	CHECK (sal > 0)
DEFAULT	Assigns a value if none is given	DEFAULT 'PUNE'

4. Cardinality and relationships

Type	Meaning	Example from notes
One to one	One parent record refers to exactly one child record	Account - Address status
One to many	One parent record refers to many child records	Department - Employee
Many to many	Records on both sides refer to many on the other side (needs a junction table)	Actor - Movie

Parent table has the primary key. **Child table** has the foreign key. Cardinality = number of unique values in a column, or the type of relationship between tables.

5. The 5 statement types

Statement	Commands	Remember
DQL (Data Query)	SELECT: projection, selection, joins	Fetch
DDL (Data Definition)	CREATE, RENAME, TRUNCATE, DROP (flashback, purge), ALTER (7 operations)	Structure, auto-commit
DML (Data Manipulation)	INSERT, UPDATE, DELETE	Data, needs COMMIT
TCL (Transaction Control)	COMMIT, ROLLBACK, SAVEPOINT	Save or undo
DCL (Data Control)	GRANT, REVOKE	Permissions

All four basic operations on a database are **CRUD**: Create/Insert, Read/Retrieve, Update/Modify, Delete/Drop.

6. Projection, selection, joins and the execution order

Projection = fetch only columns. **Selection** = fetch rows (and columns). **Joins** = fetch data from multiple tables at once. **Working of SELECT:** FROM runs first and checks the table exists (else 'table or view does not exist'). Then SELECT looks for the column (else 'invalid identifier').

7. WHERE vs HAVING, GROUP BY

WHERE	HAVING
Filters rows before grouping	Filters groups after GROUP BY
No aggregate functions	Aggregates allowed (SUM, COUNT, MAX...)
Single row filter	Group filter. Can use a sub query: HAVING SUM(sal) > (SELECT SUM(sal) FROM emp GROUP BY job...)

TIER 2: Very likely

8. Operators from your notes

Operator	Key points	Syntax / example
Logical: AND, OR, NOT	Pass multiple filter conditions in WHERE and HAVING. AND = all true. OR = any true. NOT = inverts.	WHERE deptno != 10 AND deptno != 20
IN / NOT IN	Handles multiple values on the right side. True if any value matches.	deptno IN (10,20,30)
BETWEEN / NOT BETWEEN	Range queries, inclusive, lower value first.	sal BETWEEN 1000 AND 3000
IS / IS NOT	NULL checks.	comm IS NULL
LIKE / NOT LIKE	Pattern matching. % = any number of characters. _ = exactly one character.	ename LIKE 'S%'
ESCAPE	Treats % or _ as a normal character.	LIKE '%\%%' ESCAPE '\'
REGEXP	Regular expression matching.	REGEXP_LIKE(ename,'^[AEIOU]')

9. Sub queries: three places to put them

Where	Why	Example
In WHERE / HAVING	Compare with a value or list from another query	WHERE sal > (SELECT sal FROM emp WHERE ename='ADAMS')
In SELECT	Use an aggregate with normal columns without GROUP BY	SELECT ename, (SELECT SUM(sal) FROM emp) FROM emp
In FROM (inline view)	Use an alias name in WHERE as a filter	SELECT * FROM (SELECT emp.*, sal*12 AS asalary FROM emp) WHERE asalary > 30000

Disadvantage in your notes: in a sub query we cannot fetch data from both tables. That is why joins are used when output columns come from several tables.

10. Joins

Join	Idea from notes	Syntax
Cartesian / cross	Crossing principle: every record of the first table merges with every record of the other tables. Rows = m x n.	FROM t1, t2 or t1 CROSS JOIN t2
Inner / equi	Only matching records	t1 JOIN t2 ON t1.col = t2.col
Natural	Inner join on same-named column automatically	t1 NATURAL JOIN t2
Self	Table joined with itself (employee and manager)	emp e JOIN emp m ON e.mgr = m.empno
Left / Right / Full outer	Matching plus unmatched records of one or both sides	t1 LEFT / RIGHT / FULL JOIN t2 ON ...

A join gives 3 types of records: **error record, matching record, unmatching record**. Joins are faster than sub queries. Joins are also called merge.

11. ALTER: the 7 operations and their rules

Operation	Syntax	Rule
Add column	ALTER TABLE t ADD col NUMBER [constraint];	NULL column: any table. NOT NULL column: only an empty table.
Drop column	ALTER TABLE t DROP COLUMN col;	Cannot drop a PK of a parent table until the connection is removed.
Rename column	ALTER TABLE t RENAME COLUMN old TO new;	Follow column naming rules.
Change datatype	ALTER TABLE t MODIFY col newtype;	Column must be empty to change the type.
NULL / NOT NULL	ALTER TABLE t MODIFY col type NULL | NOT NULL;	Not applicable to PK.
Add constraint	ALTER TABLE t ADD CONSTRAINT name PRIMARY KEY (col);	Only one PK per table.
Drop constraint	ALTER TABLE t DROP CONSTRAINT name;	

12. DML

```
INSERT INTO t VALUES (v1, v2, v3);                -- all columns in order
```

```
INSERT INTO t (c1, c2) VALUES (v1, v2);            -- most commonly used
UPDATE t SET c1 = v1, c2 = v2 WHERE condition;
DELETE FROM t WHERE condition;
```

13. Functions

Group	Functions	Example
Date	SYSDATE, CURRENT_DATE, SYSTIMESTAMP (with time zone), MONTHS_BETWEEN(d1,d2), ADD_MONTHS(d,n), LAST_DAY(d)	SELECT MONTHS_BETWEEN(SYSDATE, hiredate) FROM emp
Conversion	TO_CHAR(date,'format'), TO_NUMBER('string'), TO_DATE('date')	TO_CHAR(SYSDATE,'DAY')
TO_CHAR formats	YEAR, MON, MONTH, DAY, DD, D, MM, HH12:MI:SS, HH24:MI:SS	'DD-MON-YYYY'
Maths	POWER(base,raise), ABS(n)	POWER(2,3) = 8, ABS(-10) = 10

Remember: **DUAL** is the dummy table used when a function does not need a real table: SELECT SYSDATE FROM DUAL;

TIER 3: Often asked by experienced interviewers

14. Codd's rules as taught in your notes

No.	Point	What to say
1	Atomic data	Each cell must hold a single value. A cell with '107,106' makes an invalid table; use a separate column or row.
2	Schema validation	Every column of the table (schema) must be validated using datatypes and constraints.
3	Multimedia	RDBMS is not suitable for multimedia. Up to 4 GB can be handled per cell using LOB types.
4	Metadata	When multimedia is stored, the compiler detects the extra information (name, size, resolution, format). This metadata is managed by SQL as a table (metatable).
5	Distribute data	The same data can be split across tables and connected with key attributes (foreign key).

15. Attribute types (from your last pages)

Type	Meaning	Example
Key attribute (KA)	Uniquely identifies a record	Eno, Aadhaar, Mno
Non-key attribute (NKA)	Does not identify a record	Ename, Sal
Primary key attribute (PKA)	The chosen key	Eno
Non-primary key attribute (NPKA)	Key attributes that were not chosen as primary (candidate keys)	Aadhaar, Mno, Vid
Composite attribute (CA)	Made of several parts	Address
Derived attribute (DA)	Calculated from others	Age from DOB, Exp from hiredate
Single / multi valued (SVA / MVA)	One value or many values in a cell	Mno with many numbers is MVA
Foreign key attribute	Connects tables after splitting to remove redundancy	deptno in EMP

16. Functional dependency

Type	Meaning	Example
Total (full) FD	All attributes depend on only ONE key attribute	Eno -> Ename, Sal, HD, Job
Partial FD (PFD)	An attribute depends on only PART of a composite key	Eno, Dno -> Dname, Loc depends on Dno only
Transitive FD (TFD)	An attribute depends on a key attribute, which itself depends on another key attribute (indirect)	Eno -> Pin -> State -> Country

Perfect table: only total dependency, no repeated records, atomic data. **Imperfect table:** has partial or transitive dependency. Normalization converts an imperfect table into a perfect one.

17. Normalization: 1NF, 2NF, 3NF, BCNF

Memory line: 'The key (1NF), the whole key (2NF), and nothing but the key (3NF), so help me Codd.' **Definition:** dividing a bigger table into smaller tables to convert an imperfect table into a perfect table.

Form	Condition	Fix
1NF	Single value in each cell, no repeated records	Split multi-valued cells into separate rows or columns. Example: STD(S1,S2,S3 subjects) becomes SID, Sname, Sub with one subject per row.
2NF	In 1NF and no partial functional dependency	Move attributes that depend on part of the key into a new table. Example: EMP(Eno, Ename, Sal, Dno, Dname, Loc) becomes EMP(Eno, Ename, Sal, Dno FK) and DEPT(Dno PK, Dname, Loc).
3NF	In 2NF and no transitive dependency	Move the indirectly dependent attributes to a new table. Example: Eno, Pin, State, Country becomes EMP(Eno, ..., Pin) and PIN(Pin PK, State) and COUNTRY(Country, Pin FK).
BCNF (Boyce-Codd)	In 3NF and every determinant is a candidate key	Final cleanup stage. Normalization is done in 4 stages.

Why normalize? Removes redundancy, avoids update/insert/delete anomalies, saves space, keeps data consistent. **Trade-off:** more tables means more joins.

Application types and history (quick theory)

Point	Stand-alone	Web	Client-server
Internet needed	No	Yes	Yes
Database	Optional	Mandatory	Mandatory
Multiple users at same time	No	Yes	No (per notes)
Installation	Mandatory	Not mandatory	Mandatory
Examples	Camera, calendar, paint, notepad	instagram.com, facebook.com	Banking, gaming applications

History line to speak: SQL was originally called SEQUEL (Structured English Query Language), built at IBM by Donald Chamberlin and Raymond Boyce, the fathers of SQL, on the basis of E.F. Codd's relational model. It was renamed SQL (trademark reasons), and SQL is not case sensitive. **Note:** your notes credit Codd with renaming it in 1981; the usual account is the trademark reason above, so use that wording in an interview.

10 rapid-fire answers based on your notes

Question	Answer
What is a database?	A place to store information in a systematic, organised manner.
What is CRUD?	Create/Insert, Read/Retrieve, Update/Modify, Delete/Drop.
What is a cell?	Smallest unit of a table. Number of cells = rows x columns.
Row and column other names?	Row = record = tuple. Column = attribute = field.
Why VARCHAR2 over CHAR?	It frees unused memory, so no waste and better performance.
What is schema validation?	Verifying proper data enters a column using datatypes and constraints.
DDL vs DML auto-commit?	DDL is auto-commit. DML is non-auto-commit, needs COMMIT.
When use sub query in FROM?	To use an alias (like annual salary) in the WHERE filter.
Joins vs sub queries?	Joins are faster and can show columns from both tables; sub queries fetch from one table only.
Define normalization	Splitting a bigger table into smaller ones to remove partial and transitive dependency, in 1NF, 2NF, 3NF and BCNF.